

Assignment Submission Sheet

DATA 350

Responsible AI

Unit 2 Assignment: Exploring Bias and Fairness in AI through
Generative AI and Computer Vision Systems

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Introduction

Bias and fairness have become central concerns in generative AI and computer vision. These systems are now used in tasks that can shape public perception, identity, and representation. When AI creates or classifies human faces, the outputs reflect the data the system was trained on. This includes its imbalances, omissions, and cultural assumptions. The goal of this assignment was to interact with generative image models and critically examine how they depict race, gender, ethnicity, and age. To explore this, I tested two web-based generative AI systems. The first system was, “This Person Does Not Exist,” which produces photorealistic faces through GAN-based synthesis. The second system was Artbreeder, which allows users to manipulate face characteristics through adjustable sliders. I then compared my results with findings from the Gender Shades project as a research benchmark for fairness and accuracy.

This activity matters for understanding AI ethics because facial generation is not neutral and it is shaped by training data and technical optimization choices. These systems do not just create people, they construct what they think a person should be. This can reinforce social stereotypes and prejudices through invisibility or distortion of certain demographic groups. Researchers like Buolamwini and Gebru (2018) have shown that commercial facial analysis systems are significantly less accurate for darker-skinned women than for light-skinned men. This demonstrates how bias can be embedded into visual technology. The growing use of generative AI in media, hiring simulations, digital identity, and advertising increases the importance of fairness and demographic representation.

Another reason this type of evaluation matters is because generative AI systems are often treated as objective or neutral. This is even though most users never see the training process behind them. People may assume that generated faces are realistic reflections of population diversity, but these tools actually reproduce whatever demographic groups were overrepresented in the model’s dataset. If certain groups are underrepresented or inaccurately depicted, then these omissions become normalized through repeated visual exposure. By examining both a passive generative model (This Person Does Not Exist) and an adjustable one (Artbreeder), this assignment makes it possible to see whether bias comes from the dataset, the algorithmic defaults, or the user-facing controls that shape how diversity can (or cannot) be represented.

I manually generated and analyzed ten synthetic faces. I observed patterns in age, gender accuracy, and facial structure that suggest certain populations were more accurately generated than others. The next section explains how I used each tool, what patterns appeared in the generated faces, and how these patterns relate to broader bias research in computer vision. The final reflection includes ethical interpretation and recommendations for how AI systems can better prevent unfair representational defaults.

Methodology

In the assignment I wanted to explore bias in generative AI systems. I used two publicly available platforms with different design goals. “This Person Does Not Exist” is a passive system. In each image a refresh instantly produces a new photorealistic human face with no controls or sliders. I refreshed the page repeatedly until I collected ten example images. I created five intended to represent men and five intended to represent women across different age ranges and racial groups. Since there are no input controls, I had to interpret afterward whether the image matched the demographic category I was looking for. This allowed me to evaluate how well the system represents variation without user intervention.

Artbreeder, in contrast, is an interactive generative platform where users can intentionally modify features. You can modify features such as ethnicity, gender, age, skin tone, and structural facial characteristics. For this system, I experimented with slider-based controls to see how easily I could produce diversity on demand. I altered skin tone, eye and nose shape, facial structure, and age while observing which traits the system could realistically represent and which traits became distorted. My approach focused on mixing multiple “genes” to force variation, rather than simply using small incremental adjustments that would have led to near identical faces.

With both tools, I kept notes on visible differences in realism, proportionality, and representational accuracy. I also specifically recorded which demographics appeared harder for the model to render. It was harder to create older adults, darker-skinned individuals, or non-Western facial features. After collecting these examples, I compared my experiences with themes identified in the Gender Shades project, which found uneven accuracy across gender and skin tone (Buolamwini & Gebru, 2018). This methodology linked personal experimentation to established research on bias in AI-generated faces.

Observations and Analysis

Across the ten images generated from “This Person Does Not Exist,” one of the clearest patterns I observed was a tendency for faces to appear much younger than the age range I attempted to represent. This was true across the board regardless of any other factors. For example, when I tried to generate a Black male in the 35–50 age range, the system produced an individual who looked closer to someone in their early twenties. The same issue occurred when I requested a 50+ year-old white male, who appeared more like an adult in his thirties. Even when ethnicity seemed accurately rendered, aging features such as wrinkles, skin texture, and looser facial structure were noticeably absent. This suggests the model has a strong default toward youthfulness, likely shaped by training data. The data probably over represents young, highly photographed individuals. This aligns with research noting that AI models often “normalize” idealized traits when insufficient variation exists in the dataset to represent the full range of human difference (Birhane & Prabhu, 2021).

A second pattern I observed was the imbalance in visual accuracy based on gender. While some of the male images generated were reasonably realistic, the female images had a much higher rate of deformities or irregularities. In my sample, only one out of five male images had visible distortion, compared to three out of five female images. These distortions included unnatural eyes, asymmetrical facial structure, and in one case, a severely deformed hand. The fact that this happened more frequently in female images suggests that the GAN may be struggling more with the fine-grained facial geometry associated with feminine features. My interpretation is that this is not just a technical failure but also a dataset imbalance issue. The women’s faces may appear less consistently in the high-resolution facial sets used for GAN training, or the model may be smoothing features excessively when attempting to represent femininity. This aligns with Gender Shades findings that darker-skinned women had the **lowest** accuracy rates in facial recognition outcomes (Buolamwini & Gebru, 2018). This reveals that gender and visual representation bias often stack rather than occur independently.

In addition to gender-based distortion, racial and cultural representation was inconsistent. This was true even when skin tone was somewhat accurate. The most explicit example was a Latina/Hispanic representation that appeared to show the woman wearing a hairnet. While it was not an overtly hostile stereotype, it implicitly connected her identity to a particular socioeconomic role or occupation. Because a hairnet is a visual symbol often associated with food service or custodial work, its appearance introduces an embedded stereotypical bias that goes beyond inaccuracy. This stood out because it was not just the clothing choice but the fact that the system selected this as a default visual marker. This supports the idea that the model reflects occupational and class-based stereotypes present in its training data.

When ethnicity was captured somewhat accurately in facial tone or bone structure, the age misrepresentation continued across all demographic groups. The 50+ Asian woman I tried to generate looked more like someone around 30, just as the Black male intended to be in his late 30s or 40s looked younger than expected. This repeated inaccuracy of age into a narrow band of youth reflects a default aesthetic bias that favors smoothness and symmetry. These features typically are associated with younger adults. This instead of giving diverse age representation, the model collapses variation into a single optimized facial category. When age is erased or minimized, identity and life experience are visually flattened, which misrepresents entire age groups.

Artbreeder presented a different but related type of bias. It gave users more control over appearance, but the system strongly resisted non-Western feature mapping unless I pushed the sliders significantly. Lighter or European facial features persisted even when I attempted to deepen skin tone or widen nose structure. To get a face that felt genuinely non-Western, I had to over-adjust multiple settings. It was almost like fighting against the defaults. I also observed that the moment I tried to make faces older, distortions appeared much more easily. The system seemed calibrated for producing idealized, fresh, young faces. This creates what some researchers call *representational narrowness*, where the range of faces the system is comfortable generating is a compressed subset of real human diversity (Raji & Buolamwini, 2019).

The difference between the two systems is that “This Person Does Not Exist” hides its bias behind realism, while Artbreeder exposes bias during user interaction. The first tool, “This Person Does Not Exist”, you do not see the fight between input and output. You are just going to be presented with an already-optimized face. With Artbreeder, you experience firsthand how much force is required to deviate from a Eurocentric or youthful default. This shows how representation bias manifests differently depending on whether diversity is supposed to happen automatically (in GANs) or is user-driven (in adjustable synthesis systems). The outcome is consistent for certain groups. These 2 systems particularly with women, older adults, and some racial identities are less faithfully generated or require additional effort to represent accurately.

Reflection and Recommendations

In closing, my overall reflection is that the most serious issue I observed was misrepresentation. It was not through total omission of certain groups, but through the narrowing of how those groups are visually allowed to appear. The repeated trend of every demographic being rendered as much younger than intended suggests that the system is not depicting real variation in age or identity. Instead, it collapses diversity into a youth centered default. When older adults are consistently shown as young, they are not only inaccurately portrayed, they are erased from realistic representation altogether. This reinforces cultural assumptions that youthfulness is the ideal to strive for. This in turn mirrors when Buolamwini and Gebru (2018) reported that some identities receive less accurate treatment in AI systems.

The second major concern was the difference in distortion rates between genders. Three of the five women had obvious deformations while only one of the five men did. Even if this originates as a technical failure rather than intentional exclusion, it still becomes a fairness problem when it repeatedly affects the same group. The Latina woman shown with a hairnet added a different layer of bias by pairing ethnicity with an implied occupation or social class. The image, rather than being neutral, introduced a visual stereotype that carries cultural meaning.

Artbreeder raised a related but subtler concern. The concern was the need to fight against the defaults to produce non-Western features. If users must overcorrect sliders just to make a face look realistically Indian, Black, or older, then the interface itself is reinforcing a hierarchy of what identities constitute the norm. This also reflects Raji and Buolamwini's (2019) argument that representation must be built into the core modeling process, not treated as an optional layer.

In order to improve fairness, these systems should expand their training datasets to include a broader range of age groups, genders, and ethnic features. Developers should also ensure that all identities are equally easy to create rather than hidden behind extra effort. Finally, transparency indicators could inform users where models are less reliable, increasing accountability and preventing biased results from being mistaken for neutral or realistic images.

References

Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, 81, 1–15.

Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. *AAAI/ACM Conference on AI, Ethics, and Society*.

Birhane, A., & Prabhu, V. (2021). Large image datasets: A pyrrhic win for computer vision? *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Appendix



12-18 years old Latino Hispanic Male



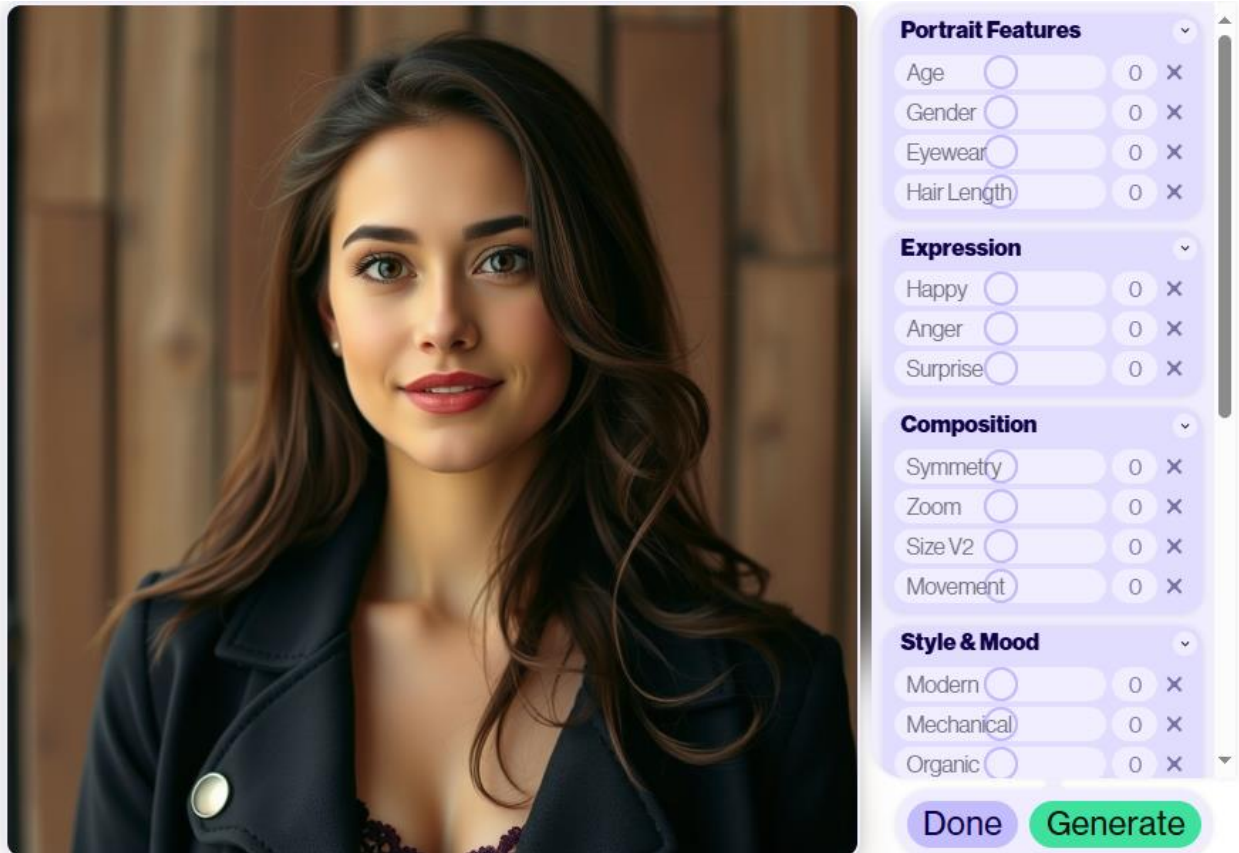
26-35 years old Indian Male



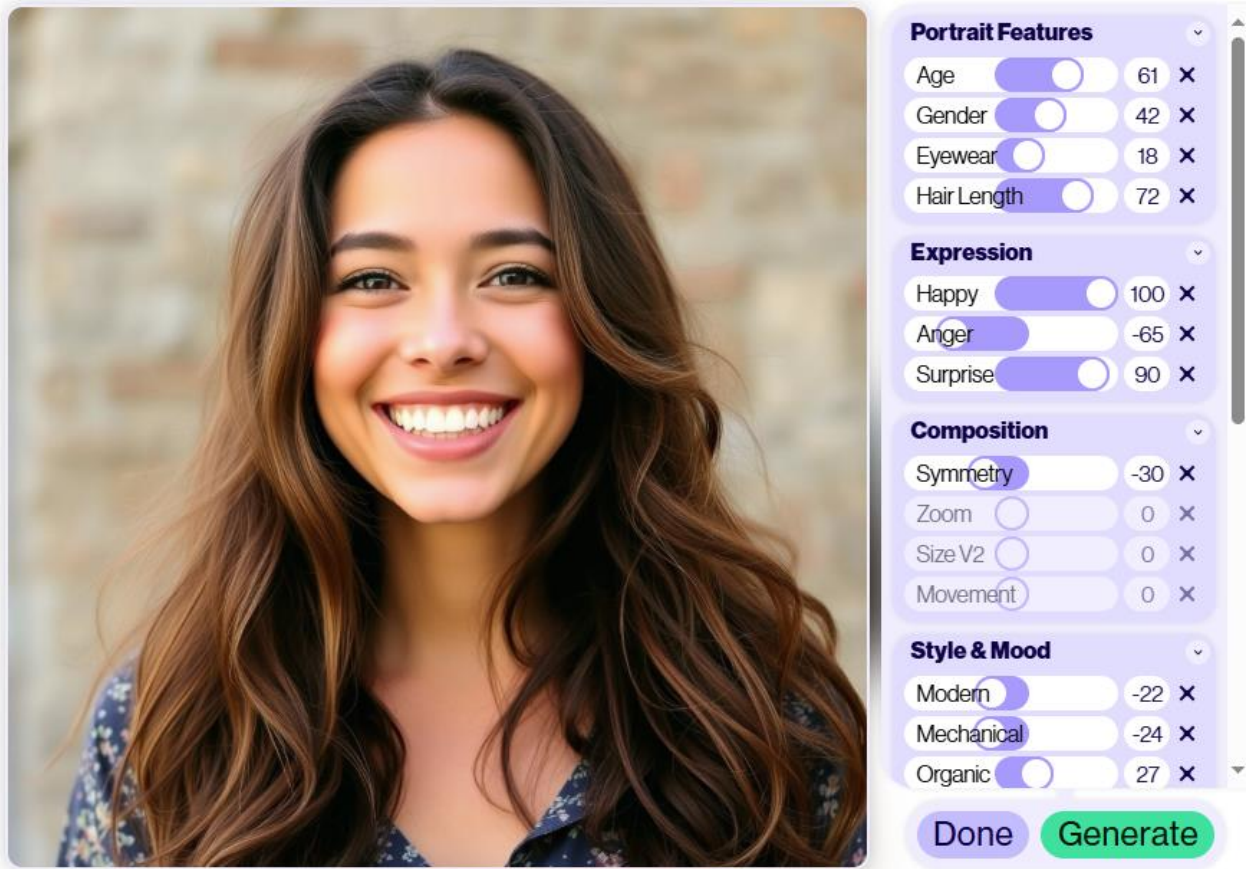
26-35 year old Indian Female



12-18 year old White Female



I asked to create a young brunette woman.



Same Image Adjusted